

Philosophy of emergence

Session 1
03.05.2024

Two questions

- What is an “emergent phenomenon”?

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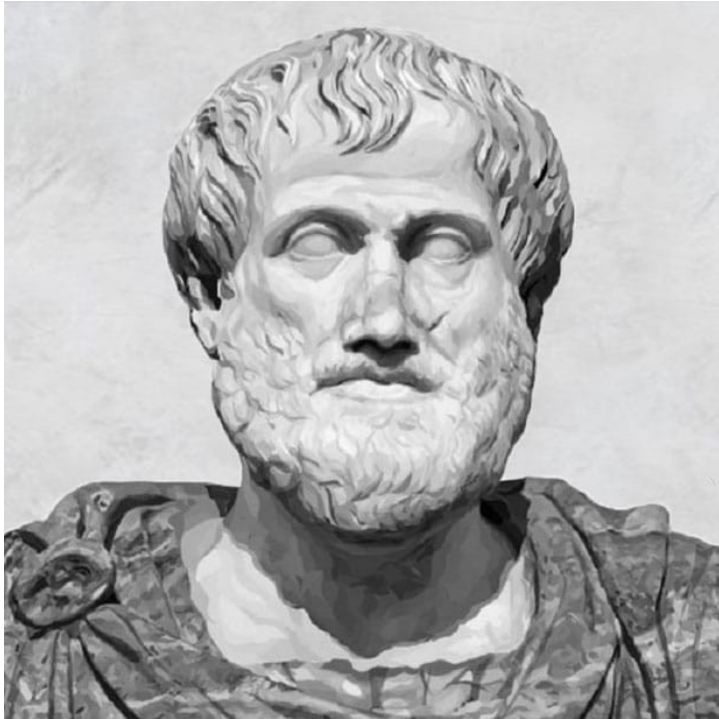
- What emergent phenomena do you know?

Background

Emergentism as a middle way between:

Substance dualism	Reductionism
“Mind and matter are not the same thing”	“Mind is <i>nothing more than</i> matter” (<i>physicalist reductionism</i>), or “Matter is <i>nothing more than</i> mind” (<i>panpsychism</i> , less common)

Background



<https://ethics.org.au/>

- Debates around emergence are far from recent!
- Aristotle: form vs. matter (*hylomorphism*). Matter is what entities/organisms are made of, form is the way this matter is organized. Body and soul are a matter/form couple.

Background

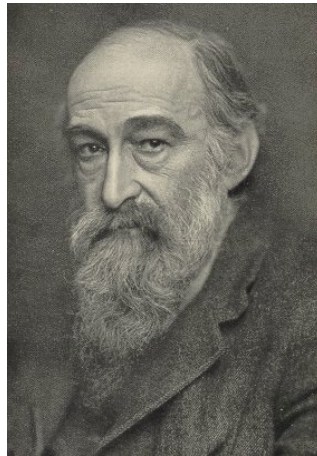
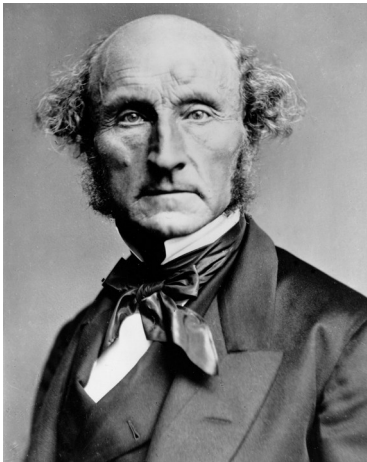
- Descartes and the Scientific Revolution (17th century): mechanistic account of the material world (but mind-body dualism)



<https://www.worldhistory.org/>

Background

- The British Emergentists (19th – 20th centuries): John Stuart Mill, Samuel Alexander, Charlie Dunbar Broad. A 'middle-way' between mechanists and vitalists
 - Vitalists: life is a substance of its own
 - Mechanists: life is nothing more than a bunch of physico-chemical processes



<https://de.wikipedia.org>

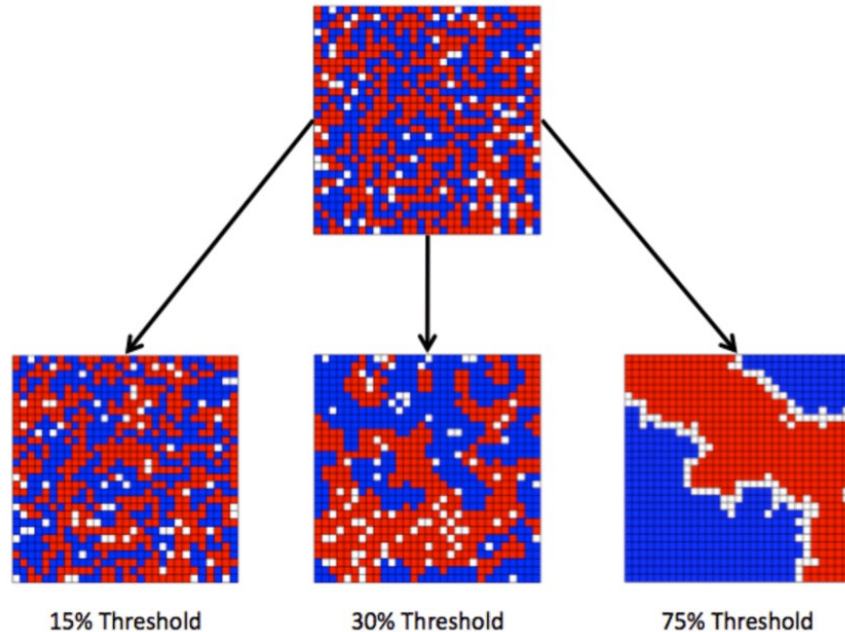
Background

- Broad's definition of emergence:

“the characteristic properties of the whole $R(A, B, C)$ [where R marks their structural arrangement] cannot, even in theory, be deduced from the most complete knowledge of the properties of A , B , and C in isolation or in other wholes which are not of the form $R(A, B, C)$.”

- Technical point: **ontology** vs. **epistemology**

Background



- 20th century: scientific advances make emergentism less appealing (quantum chemistry, molecular biochemistry)
- 1970s: developments and discoveries in complex systems science: emergence is back!

Visualization of Thomas Schelling's (1971) segregation model.

Jackson, J. C., Rand, D., Lewis, K., Norton, M. I., & Gray, K. (2017). Agent-based modeling: A guide for social psychologists. *Social Psychological and Personality Science*, 8(4), 387-395.

O'Connor, T. (2020). Emergent properties. *Stanford Encyclopedia of Philosophy*

Background

- Today: different branches and definitions of emergence
 - For complex systems, mostly “weak” (= compatible with causal closure) or “epistemological” varieties.
 - Strong emergentism not very popular, but sometimes put forth to account for philosophically-loaded notions such as consciousness or free will
... or phenomena within physics (quantum entanglement, fractional quantum Hall effect, spacetime...), but this is quite niche.

Background

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Why talk about emergence?

- Emergence has many definitions; good to know in what sense certain phenomena can be called 'emergent'
- Good to know how to react in 'ideological' debates
- Emergence can be a useful lens to conceptualize scientific and philosophical problems (tension between 'dependence' and 'novelty', the notion of causation, etc.)

Foundations of emergence

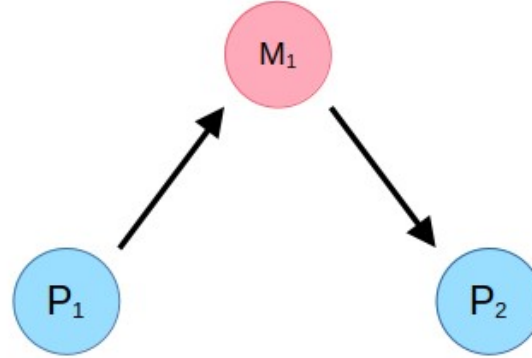
The mind-body problem

Are mental entities (consciousness, intentions, sensations, etc.) the same as physical entities (the brain, action potentials, C-fiber activation, etc.)?

Two major answers:

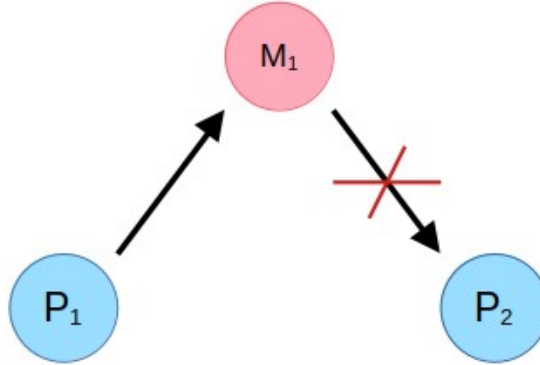
Foundations of emergence

Substance dualism (Descartes, Leibniz): mind and matter are **not** the same thing



Foundations of emergence

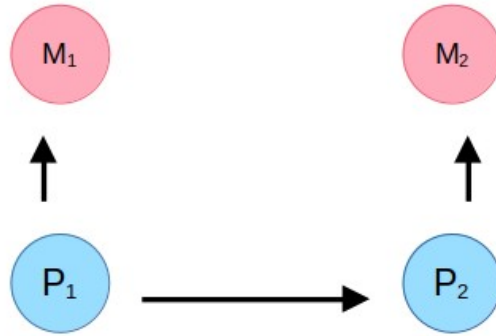
Substance dualism (Descartes, Leibniz): mind and matter are **not** the same thing



Problem: argument of the **causal closure of the physical**: physical causes are fully sufficient to explain physical effects. Human behavior is a physical effect – therefore, no mental entities are needed to explain it.

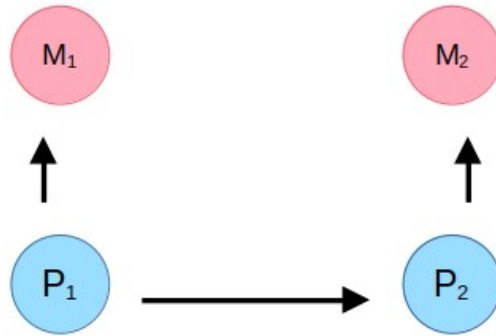
Foundations of emergence

Epiphenomenalism: mental phenomena have no causal power on physical phenomena



Foundations of emergence

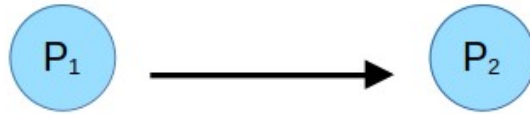
Epiphenomenalism: mental phenomena have no causal power on physical phenomena



Problem: Yet one would like to say that mental phenomena such as intentions, beliefs, etc., influence behavior

Foundations of emergence

Ontological physicalism: mental entities are physical entities



Foundations of emergence

Ontological physicalism has become predominant. But mind-body problem has not disappeared: how to account for intentions, beliefs, consciousness, in physical terms? Is such an account possible?

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→ Reduction problem: can mental phenomena be reduced to physical phenomena? Is pain *nothing more than* C-fiber activation?

Foundations of emergence

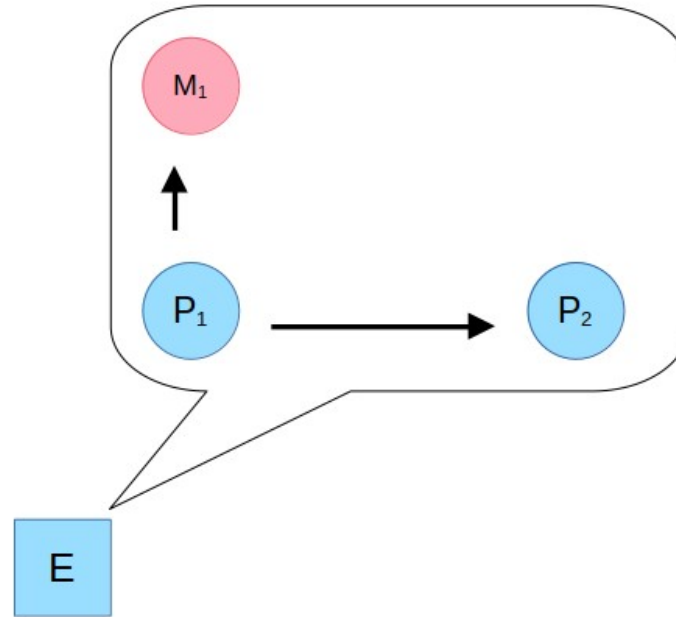
Reductive physicalism: Mental phenomena can be reduced to physical phenomena.

But how?

Huge literature and many programs to “naturalize the mind”

Foundations of emergence

Nonreductive physicalism: mental properties cannot be reduced to physical properties (*property dualism*)



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Foundations of emergence

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Note: entails special science autonomy; but special science autonomy does not entail emergentism (in an ontological sense)

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Note: nonreducible domains of properties can be within the same special science (e.g., physics; cf. Paul Humphreys, Robert Batterman)

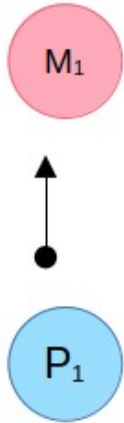
Supervenient emergentism

Property dualism is vague: how do mental and physical properties interact?

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Mental properties **supervene** on physical properties: mental properties depend on physical properties and are determined by them

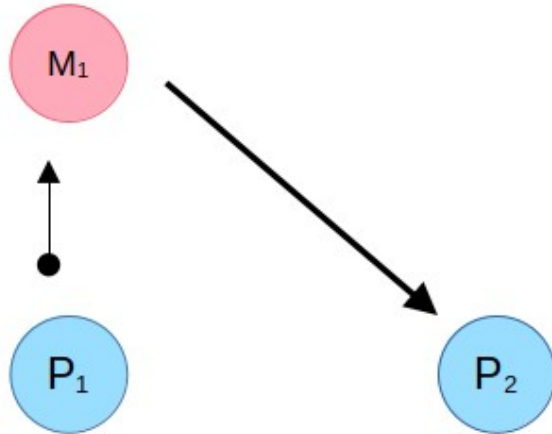


Supervenient emergentism

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Mental states can **cause** physical states



Supervenient emergentism

Differences between supervenience and causation?

Supervenient emergentism

Differences between supervenience and causation?

Supervenience: Comes from moral philosophy

Once you fix the physical, you fix the mental

or: any two entities that have the same physical properties will have the same mental properties. = some kind of covariation

But we do not necessarily know in virtue of what (i.e., what the mechanism behind that is)

Supervenient emergentism

Differences between supervenience and causation?

Causation: has many definitions (counterfactual accounts, probabilistic accounts...) but can be mobilized in explanations

Supervenience is trivially reflexive, not causation

Supervenient emergentism

Nonreductive physicalism – 4 theses:

1. *Ontological physicalism*. The world is exclusively composed of aggregates of material particles
2. *Irreducibility of the mental to the physical*. Mental properties are not reducible to physical properties.
3. *Mind-body supervenience*. Mental properties supervene on physical properties.
4. *Causal efficacy of the mental*. Mental events are causally efficacious, i.e., they can cause other (mental or physical) events.

Supervenient emergentism

Supervenient emergentism – 4 theses:

1. *Emergence*. When aggregates of material particles come to an adequate complexity level, authentically new properties emerge and characterize these systems.
2. *Irreducibility of emergent properties*. Emergent properties are not reducible to their “base conditions”, i.e., the underlying conditions they emerge from.
3. *Supervenience of emergents on their base conditions*. Emergent properties supervene on their base conditions.
4. *Downward causation*. Emergent properties possess causal powers to influence phenomena at the level they emerged from.

Kim's criticisms against supervenient emergentism



1st criticism: the need for a positive
characterization of emergence

Jaegwon Kim (1934 - 2019)

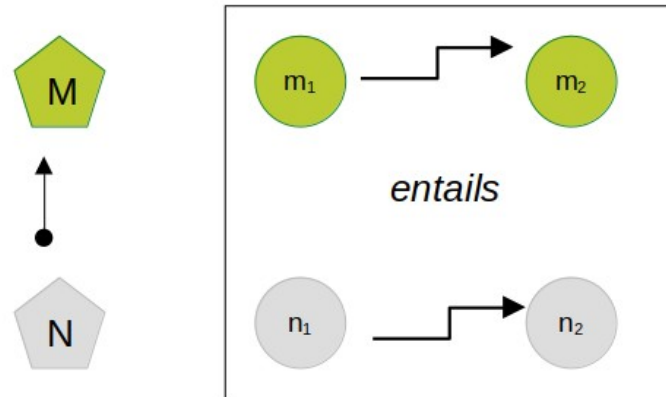
<https://www.francoisloth.com/>

Kim's 1st criticism

Definition of **strong supervenience**:

M (strongly) supervenes on N iff for all property m in M, for all x that instantiates m , there is in N a property n such that x instantiates n and, necessarily, if something instantiates n , it instantiates m

“translation”: there cannot be an M-difference without an N-difference



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Kim's criticism: this definition is too vague – just means there is a covariation between two domains of properties, and why this covariation occurs is not explained (for emergentists, it is unexplainable *in principle*).

Kim's 1st criticism

Irreducibility. Which reduction are we talking about? Examples:

“Nagelian” reduction: a theory T1 is reducible to a theory T2 iff one can deduce T1's laws and descriptions from T2's + set of adequate corrections and definitions brought to T2.

Kim's 1st criticism

Irreducibility. Which reduction are we talking about? Examples:

Reduction by identity: an entity domain is reducible by identity to another domain iff each element from the first domain can be identified to an element of the second.

Kim's 1st criticism

Irreducibility. Which reduction are we talking about? Examples:

Functional reduction: a property F is functionally reducible iff one can identify a physical property P that fills F's functional role, where F's functional role is the causal task(s) F's realizer(s) play in a given organism.

Kim's 1st criticism

Irreducibility. Which reduction are we talking about? Examples:

Kim's criticism: Emergentists do not specify what type of reduction they target. Different types of reduction can lead to different types of emergentism.

Kim's 2nd criticism: Downward causation argument

Argument in two steps: **exclusion argument** and **supervenience argument**

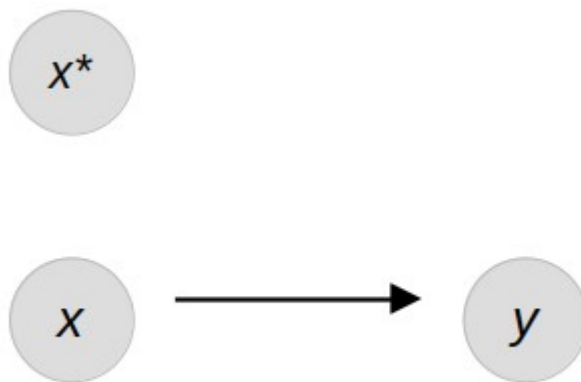
Kim's 2nd criticism: Downward causation argument

Exclusion argument

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Exclusion argument

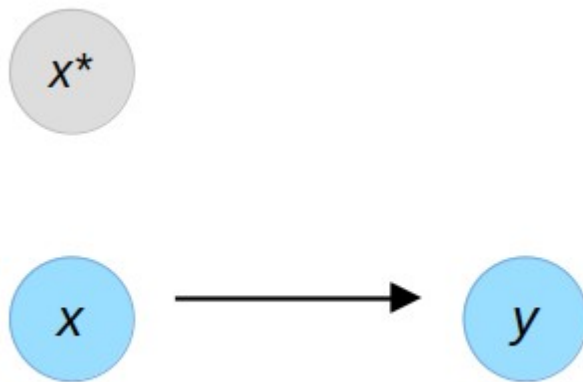
Causal exclusion. If an event x is causally sufficient for another event y , then no event x^* different from x is causally relevant for y .



Kim's 2nd criticism: Downward causation argument

Exclusion argument

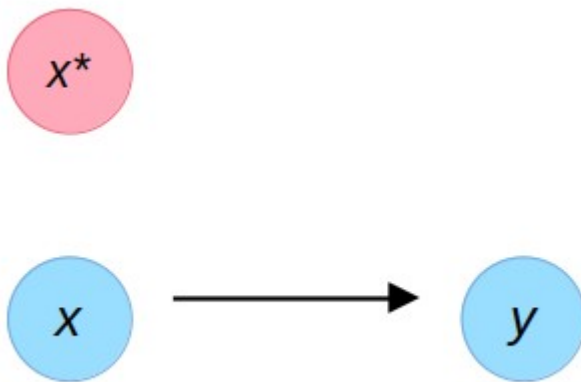
Causal closure of the physical. For all physical events y , there is a physical event x that is causally sufficient for y .



Kim's 2nd criticism: Downward causation argument

Exclusion argument

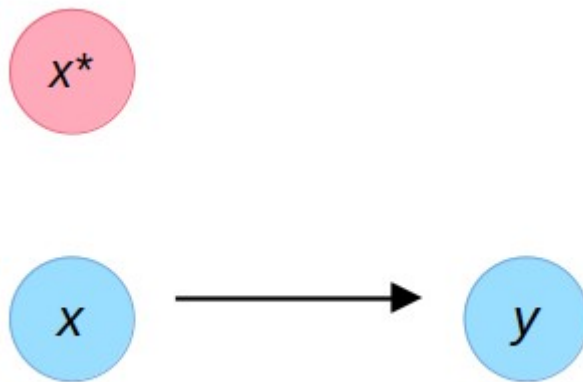
Property dualism. For all physical events x and all mental events x^* , x is distinct from x^* .



Kim's 2nd criticism: Downward causation argument

Exclusion argument

Conclusion (epiphenomenalism). For all physical events y , no mental events x^* are causally relevant for y .



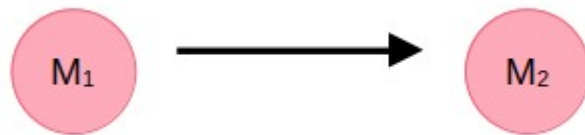
Kim's 2nd criticism: Downward causation argument

Supervenience argument

Demonstrates that any doctrine that accepts supervenience has to accept downward causation

Kim's 2nd criticism: Downward causation argument

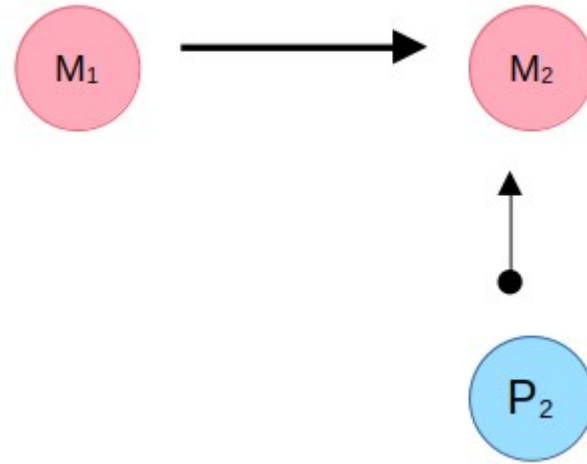
Supervenience argument



Assume a mental event M1 causes another mental event, M2

Kim's 2nd criticism: Downward causation argument

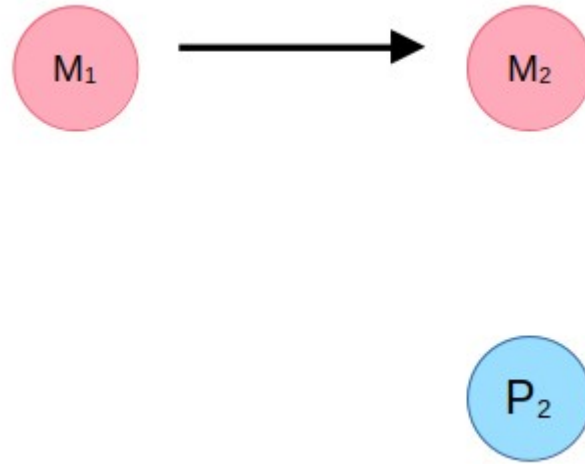
Supervenience argument



M2 is emergent, so its base condition P2 is instantiated at the same time as M2.

Kim's 2nd criticism: Downward causation argument

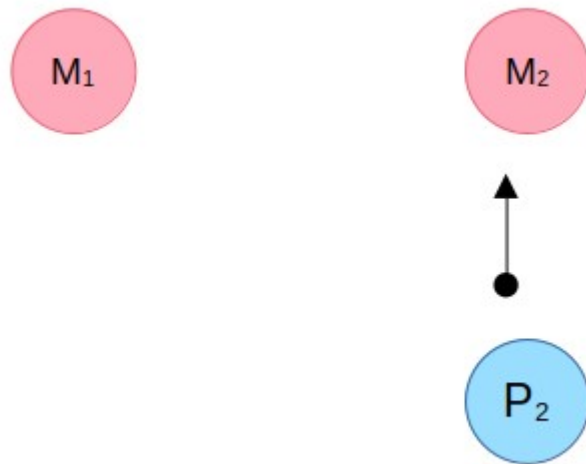
Supervenience argument



Therefore, M2 can be explained in two ways: it appeared because M1 caused it,

Kim's 2nd criticism: Downward causation argument

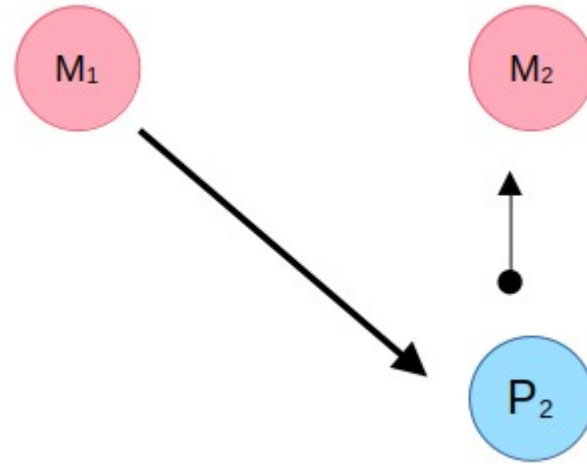
Supervenience argument



or it appeared because P_2 instantiated it

Kim's 2nd criticism: Downward causation argument

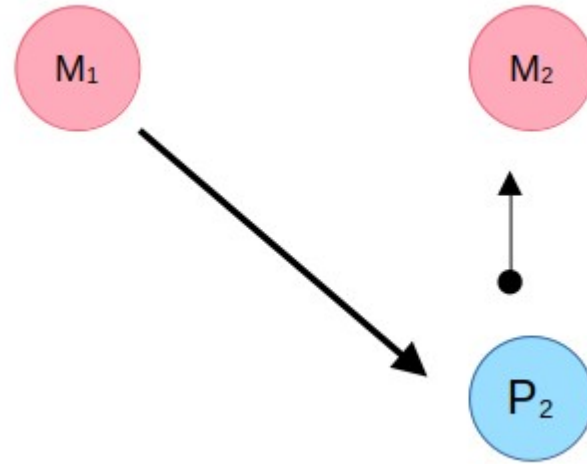
Supervenience argument



But each of M_1 and P_2 are on their own sufficient to explain M_2 . To avoid overdetermination, the only solution is to assert that M_1 caused M_2 by causing its base P_2

Kim's 2nd criticism: Downward causation argument

Supervenience argument



This is downward causation. Therefore, any doctrine that accepts supervenience has to accept downward causation

Kim's 2nd criticism: Downward causation argument

The exclusion argument shows that downward causation is impossible, and the supervenience argument, that emergentists have to accept downward causation. Therefore, Kim's arguments demonstrate that emergentism is inconsistent.

Consequences for emergentists

Emergentists have to reject either one of their four theses, or one of the premises in Kim's argument.

Rejecting downward causation is problematic, as emergentists want to avoid epiphenomenalism.

Rejecting irreducibility is problematic as well – otherwise, emergentists would become reductionists.

Some authors reject the causal closure of the physical, others, the exclusion principle.

Thank you for your attention!

Literature

Batterman, R. W. (2001). The devil in the details: Asymptotic reasoning in explanation, reduction, and emergence. Oxford University Press.

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